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Course title: Design and Analysis of Algorithms

CO3 _ASSESSMENT TOOL 2 _Case Study Based Assessment

Case Study: Web Server Log Analysis (Sorting + Binary Search)

Aim:

To analyze web server log files efficiently by combining sorting algorithms with Binary Search for faster searching.

Problem Statement:

- A web server generates thousands of log records every day, resulting in large datasets.
- Log records are often stored in an unsorted format, making data retrieval inefficient.
- Searching for a specific log entry requires scanning the entire file, which increases processing time.
- As the log size grows, search performance decreases and system resource usage increases.
- Slow log analysis delays troubleshooting, security monitoring, and performance evaluation.
- An efficient sorting and searching solution is required to reduce search time, improve performance, and handle large volumes of log data effectively.

Issues Identified:

- Large amount of log data.
- Slow search in unsorted logs.
- High processing time.
- Increased memory usage for large datasets.

Algorithm:

- **Start.**
- Read the web server log records from the log file.
- Apply **Merge Sort** to divide the log records into smaller parts, sort them, and merge them into a single sorted log file.
- Store the sorted log records.
- Enter the log entry (timestamp, IP address, or request) to search.
- Apply **Binary Search** on the sorted log records.
- Compare the middle record with the search key.
- If the record matches, display the complete log entry.
- Otherwise, continue searching in the left or right half until the record is found or the search range becomes empty.
- If no matching record is found, display "**Log Entry Not Found.**"
- **Stop.**

Solution:

- **Collect the web server log data:** Gather log records containing details such as timestamp, IP address, request, and status code.
- **Sort the log entries using Merge Sort:** Organize the log records in ascending order using the Merge Sort (Divide and Conquer) algorithm.
- **Store the sorted log file:** Save the sorted records to enable faster and more efficient searching.
- **Search using Binary Search:** Apply the Binary Search algorithm to quickly locate the required log entry in the sorted file.
- **Display the matching record:** Show the complete details of the matching log entry or display "**Log entry not found**" if no match exists.

Divide and Conquer Technique:

- Merge Sort divides the log file into smaller parts, sorts each part, and merges them.

- Binary Search repeatedly divides the sorted data into two halves until the target record is found.

Complexity Analysis:

- Merge Sort: $O(n \log n)$
- Binary Search: $O(\log n)$
- Overall Complexity: $O(n \log n)$

Efficiency Justification:

Sorting the log file once allows Binary Search to locate records much faster than Linear Search. This approach reduces search time, improves performance, and is suitable for handling very large web server log files.

Result:

The combination of Merge Sort and Binary Search provides an efficient solution for web server log analysis by reducing processing time and enabling fast searching in large datasets.